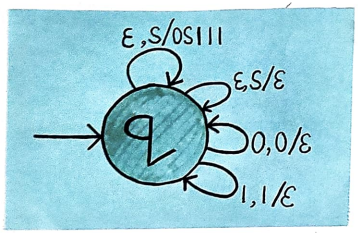
PDA : State, Diagram & Traces

Q1. Convert $S \to 0S111 \mid \varepsilon$ into PDA and test 00111111



RULE	TRANSITION
$S \to 0S111$	$\delta(q, \varepsilon, S) = (q, 0S111)$
$S \to \varepsilon$	$\delta(q, \varepsilon, S) = (q, \varepsilon)$
Match 0	$\delta(q, 0, 0) = (q, \varepsilon)$
Match 1	$\delta(q, 1, 1) = (q, \varepsilon)$

Accept by empty stack
$L = \{0^n 1^{3n}\}$

→ Testing 00111111

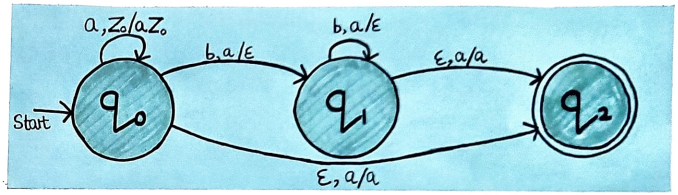
STEP	STATE	INPUT LEFT	STACK	MOVE
1	q	00111111	S	start
2	q	00111111	0S111	$S \to 0S111$
3	q	0111111	S111	pop 0
4	q	0111111	0S11111	$S \to 0S111$
5	q	111111	S11111	pop 0
6	q	111111	111111	$S \to \varepsilon$
7	q	11111	11111	pop 1
8	q	1111	1111	pop 1
9	q	111	111	pop 1
10	q	11	11	pop 1
11	q	1	1	pop 1
12	q	ε	ε	pop 1

Accepted!

INPUT & STACK are both empty so 00111111 is accepted (n=2, six 1's)

Method used : One state q. Each production becomes an ε-move that replaces the variable on the top of the stack. Each Terminal gets a move that pops it when the input matches. This accepts on empty stack.

Q2. Construct a PDA for $L = \{a^i b^j \mid i > j\}$ and test the strings aaabb and aabbb



TRANSITION	MEANING
$\delta(q_0, a, Z_0) = (q_0, aZ_0)$	push first a
$\delta(q_0, a, a) = (q_0, aa)$	push more a's
$\delta(q_0, b, a) = (q_1, \varepsilon)$	first b, pop an a
$\delta(q_1, b, a) = (q_1, \varepsilon)$	more b's, pop an a
$\delta(q_0, \varepsilon, a) = (q_2, a)$	no b's, a's left over
$\delta(q_1, \varepsilon, a) = (q_2, a)$	b's matched, extra a left

Accept by Final state start stack at Z_0!

Idea: push a for each a, pop one for each b. Accept if atleast one a is still left, which means $i > j$. Z_0 is the bottom marker, if Z_0 is on top there are no a's left.

→ TEST : aaabb

STEP	STATE	INPUT LEFT	STATE
1	q_0	aaabb	Z_0
2	q_0	aabb	aZ_0
3	q_0	abb	aaZ_0
4	q_0	bb	$aaaZ_0$
5	q_1	b	aaZ_0
6	q_1	ε	aZ_0
7	q_2	ε	aZ_0

Accepted!

INPUT Consumed and PDA is in FINAL state q_2 so "aaabb" is accepted! (3a's > 2b's)

→ TEST : aabbb

STEP	STATE	INPUT LEFT	STATE
1	q_0	aabbb	Z_0
2	q_0	abbb	aZ_0
3	q_0	bbb	aaZ_0
4	q_1	bb	aZ_0
5	q_1	b	Z_0
6	Stuck	b	Z_0

Rejected!

No move from b when Z_0 is on Top, so "aabbb" is rejected!

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